

Sequences Worksheet

In problems 1-17, an explicit formula for a_n is given. Determine if the sequence converges or diverges.

If it converges, find $\lim_{n \rightarrow \infty} a_n$.

1. $a_n = \frac{n}{2n-1}$ C. to $\frac{1}{2}$

3. $a_n = \frac{4n^2 + 1}{n^2 - 2n + 3}$ C. to 4

5. $a_n = \frac{n+4}{2n^2+1}$ C. to 0

7. $a_n = (-1)^n \frac{n}{n+1}$ D. oscillates between -1 & 1

9. $a_n = \frac{\sin(n\pi/2)}{n}$

11. $a_n = \frac{e^n}{n^2}$ $e^n \leftarrow$ exponential (faster)
 $n^2 \leftarrow$ poly. $\lim_{n \rightarrow \infty} \frac{e^n}{n^2} = \infty$ diverges

13. $a_n = \frac{(-\pi)^n}{4^n}$ C. to 0
 $\leftarrow$ faster

15. $a_n = 1 + (0.9)^n$ C. to 1

17. $a_n = \frac{\ln n}{n}$ C. to 0
 $\leftarrow$ faster

2. $a_n = \frac{3n+1}{n+2} \rightarrow 3$

4. $a_n = \frac{3n^2+2}{n+4}$ D. to ∞

6. $a_n = \frac{\sqrt{n}}{n+1}$ C. to 0

8. $a_n = \frac{n \sin(n\pi/2)}{2n+1}$

10. $a_n = e^{-n} \cos n = \frac{\cos n}{e^n}$ C. to 0

12. $a_n = \frac{e^n}{2^n}$ $\lim_{n \rightarrow \infty} \frac{e^n}{2^n} \leftarrow$ faster
 ∞

14. $a_n = \left(\frac{1}{2}\right)^n + 2^n$ C. to 0 D. to ∞

16. $a_n = \frac{n^3}{e^n}$ C. to 0



n^n

factorial $n!$

exponential c^n

polynomial or root n^c

logs

$\log n$



C onstants

n	$n!$
1	1
2	2
3	6
4	24
5	120
6	720
7	5040

$n! = n(n-1)(n-2) \cdot 3 \cdot 2 \cdot 1$

Sequences

Tell if each sequence converges or diverges.
If it converges find the limit.

8 | 40320

$$(1) a_n = \frac{n^2 + 3n - 1}{2n^2 - 5n} \quad \frac{1}{2}$$

(6)

$$a_n = \frac{n!}{10^n}$$

$$(2) a_n = \frac{3^n + n^4}{2^n + n^7} \quad D \rightarrow \infty$$

(7)

$$a_n = \frac{(n+1)!}{(n+2)!}$$

$$(3) a_n = \frac{4n+5}{1,001^n} \quad 0$$

(8)

$$a_n = \frac{3}{n+4}$$

$$(4) a_n = \frac{n^{100,000}}{1,000,001^n} \quad 0$$

(9)

$$a_n = \sin n$$

$$(5) a_n = \frac{e^n + 5}{2 - 3^n} \quad 0$$

(10)

$$a_n = \frac{\sin n}{n}$$

$$(6) a_n = \frac{n!}{10^n} \quad \infty$$

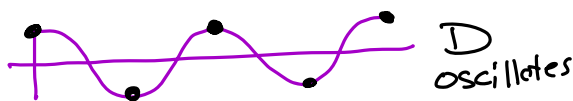
$$(7) a_n = \frac{(n+1)!}{(n+2)!} = \frac{\cancel{(n+1)} n!}{(n+2) \cancel{(n+1)} n!} = \frac{1}{n+2} \quad 0$$

$$(8) a_n = \frac{3}{n+4} \quad 0$$

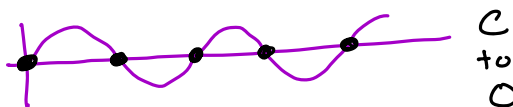
$$(9) a_n = \sin n \quad D. \text{ (erotic)}$$

$$(10) a_n = \frac{\sin n}{n} \quad C. 0$$

$$(11) a_n = \cos(\pi n)$$



$$(12) a_n = \sin(\pi n)$$



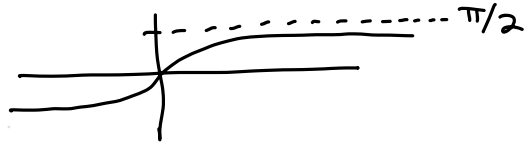
Sequences

Tell if each sequence converges or diverges.
If it converges find the limit.

⑬ $a_n = \cos\left(\frac{\pi}{2}n\right)$ D

⑭ $a_n = \tan n$ D

⑮ $a_n = \tan^{-1} n$ C to $\frac{\pi}{2}$



⑯ $a_n = \frac{\sqrt{n^2 + 9}}{n} = 1$

11.2 Infinite Series

- If you add up the terms of a sequence, you get a **series**.

$$\sum_{i=1}^{\infty} a_i = a_1 + a_2 + a_3 + a_4 + a_5 + \cdots$$

- If you only add up the first n terms of a sequence, you get the n^{th} **partial sum** of the series, denoted by s_n .

$$s_n = a_1 + a_2 + a_3 + a_4 + \cdots + a_n$$

0.0.1 Example:

Consider the series $\sum_{n=1}^{\infty} \frac{1}{2^n}$

- Write the first four terms of the series.

$n:$ 1 2 3 4

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$$

- Compute the following partial sums:

$$\bullet s_1 = \frac{1}{2}$$

$$s_5 = \frac{31}{32}$$

$$\bullet s_2 = \frac{3}{4}$$

$$s_6 = \frac{63}{64}$$

$\vdots$

$$\bullet s_3 = \frac{7}{8}$$

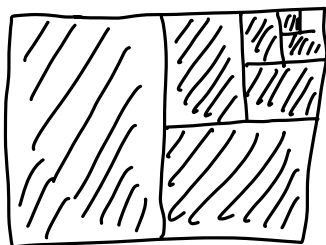
$$s_n = \frac{2^n - 1}{2^n}$$

$$\bullet s_4 = \frac{15}{16}$$

$$\lim_{n \rightarrow \infty} \frac{2^n - 1}{2^n} = 1$$

- If the sequence of partial sums converges, then we say that the series converges.

$$\sum_{n=1}^{\infty} \frac{1}{2^n} \text{ converges to } \underline{1}$$



With some series, it's easy to tell if they converge or diverge:

0.0.2 Example:

Write out a few terms or partial sums and determine whether the series converges or diverges.

$$1. \sum_{n=1}^{\infty} 0 = 0 + 0 + 0 + 0 + \dots$$

Converges to 0

$$4. \sum_{n=1}^{\infty} (-1)^n = -1 + 1 - 1 + 1 - 1 + \dots$$

partial sums: $-1, 0, -1, 0, \dots$
D. by oscillating.

$$2. \sum_{n=1}^{\infty} n = 1 + 2 + 3 + 4 + \dots$$

D. to ∞
partial sums: $1, 3, 6, 10, \dots$

$$5. \sum_{n=1}^{\infty} \frac{n}{n+1}$$
$$= \frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \frac{5}{6} + \dots$$

D to ∞

$$3. \sum_{n=1}^{\infty} 1 = 1 + 1 + 1 + 1 + \dots$$

D. to ∞

Note: the terms converge to 1,
but the sum diverges to ∞

But some require some analysis to determine if they converge or diverge.

0.0.3 Example:

Examine these series and guess whether they converge or diverge. (Maybe use a spreadsheet)

$$1. \sum_{n=1}^{\infty} \frac{1}{n} \quad \text{D}$$

$$3. \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} \quad \text{D}$$

$$2. \sum_{n=1}^{\infty} \frac{1}{n^2} \quad \text{C}$$

$$4. \sum_{n=1}^{\infty} \frac{1}{n!} \quad \text{C to } e-1$$

↑

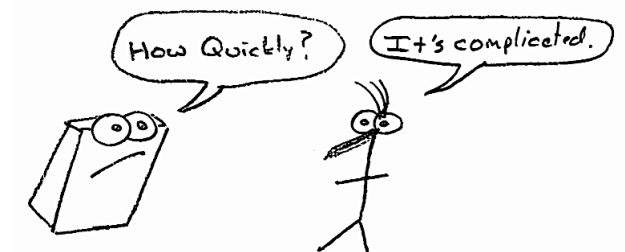
Big Questions:

1. Given a series, does it converge or diverge?
2. If it converges, what is the sum?

Partial Answer:

In order for a series to converge, the terms must:

- Shrink to 0, and
- They must shrink to 0 quickly.



1 The Test for Divergence

Test For Divergence (TFD)

If $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum a_n$ diverges.

Restated: If the terms don't go to 0, the series diverges.

Important notes:

- If the terms DO go to 0, then the TFD fails (i.e. gives no information).
- The TFD **never** tells you that a series converges.

1.0.1 Example:

What does the TFD tell us about these series?

(a) $\sum_{n=1}^{\infty} \frac{2n+1}{3n+\sqrt{n}}$

$\lim_{n \rightarrow \infty} \frac{2n+1}{3n+\sqrt{n}} = \frac{2}{3} \therefore \text{Series will diverge}$

(b) $\sum_{n=1}^{\infty} \left[1 + \left(\frac{1}{2} \right)^n \right]$

$\lim_{n \rightarrow \infty} \left(1 + \underbrace{\left(\frac{1}{2} \right)^n}_0 \right) = 1 \therefore \text{Diverge}$

(c) $\sum_{n=1}^{\infty} \frac{1}{n}$ $\lim_{n \rightarrow \infty} \frac{1}{n} = 0 \leftarrow$ TFD is inconclusive.

(d) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ $\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0 \leftarrow$ TFD is inconclusive.

2 Geometric Series

In a geometric series, there is a common ratio between any two adjacent terms. The common ratio is called r and the first term is a .

2.0.1 Example:

Find a and r for each series:

(a) $1 + 2 + 4 + 8 + \dots$ $a=1$ $r=2$ Diverge

(b) $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$ $a=\frac{1}{2}$ $r=\frac{1}{2}$ Converge $\text{sum} = \frac{\frac{1}{2}}{1 - \frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$

(c) $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$ $a=1$ $r=\frac{1}{3}$ Converge $\text{sum} = \frac{1}{1 - \frac{1}{3}} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$

(d) $12 - 3 + \frac{3}{4} - \frac{3}{16} + \dots$ $a=12$ $r=-\frac{1}{4}$ Converge $\text{sum} = \frac{12}{1 - (-\frac{1}{4})}$

Geometric Series

$$\sum_{n=1}^{\infty} ar^{n-1}$$

$$= \frac{12}{\frac{5}{4}} = \frac{12 \cdot 4}{5} = \frac{48}{5}$$

- A geometric series converges if and only if $|r| < 1$.
- If it converges, the sum is $\frac{a}{1-r}$